M2 Progress Report - Team 254

Kew: Green Home Score and Retrofit Recommendation Engine

1. Current approach and implementation updates

Kew assesses a home's energy performance, identifies suitable retrofits, and estimates their energy, carbon and potential property-value impacts. The implementation separates numerical calculations from language generation: backend engines produce every estimate, while the LLM explains verified outputs.

The current pipeline includes:

- Green Home Score: deterministic weighted scoring across the seven required components, with documented mappings, contribution-level explanations and explicit missing-data handling.

- Retrofit recommendations: property-specific feasibility checks, High/Medium/Low suitability, individual retrofit simulations and prioritised recommendations.

- Package simulation and optimisation: evaluation of up to 32 combinations of the five supported upgrades. Package effects account for interactions rather than adding independent savings. Optimisation supports carbon reduction, energy reduction and shortest payback within a stated budget.

- Energy and carbon estimates: separate electricity and gas changes, with negative savings representing increased consumption. Carbon calculations use documented per-fuel emissions factors.

- Property-value estimation: a guarded regression pipeline that compares consistent before/after property states. Critical missing inputs produce an explicit unavailable response.

- AI advisor: local Ollama generates conversational explanations from verified evidence. What-if requests trigger backend calculations before explanation. Numerical sentences are rendered by the server to preserve units, signs and before/after meaning.

Key implementation updates include an integrated synthetic property-value impact model, broader held-out evaluation, clearer recommendation tradeoffs, suitability badges, live advisor progress and a reproducible submission workflow. The chatbot now interprets results in natural language instead of displaying raw fact lists. Customer memory remains outside the current demo scope.

2. Experiments and results

Property-value model

A documented generator creates 1,800 synthetic property/scenario pairs. Properties are separated into fitting, model-selection, calibration and held-out test partitions, preventing the same property from appearing across partitions.

Ridge regression was compared with HistGradientBoostingRegressor. The latter was selected using validation performance and constrained so improved energy intensity cannot reduce predicted value.

Held-out metric	Result
Selected model price MAE	£23,642
Ridge baseline price MAE	£67,422
Selected model price RMSE	£32,866
Selected model price R^2	0.992
Property-value change MAE	£5,655
Price interval coverage	90.8%
Property-value change interval coverage	86.4%

The interval target is nominally 90%; the observed change-interval coverage remains below that target and is disclosed. These results measure agreement with documented synthetic relationships, not real-world market accuracy or causal retrofit effects.

3. Progress on testing

Held-out energy and recommendation evaluation

A separate evaluation set contains 24 synthetic homes covering different property types, heating systems, existing upgrades and eligibility conditions. The harness examines all 32 packages per home and uses a separately implemented analytical reference.

- 119 feasible package cases: per-fuel energy and carbon calculations agreed with reference estimates within numerical tolerance.

- 120 feasibility cases: classifications matched the documented reference rules.

- 72 ranking and 72 optimisation cases: results matched the reference policy across the three objectives.

These checks establish consistency with the documented assumptions. Because the reference shares those assumptions, they do not demonstrate measured real-world savings accuracy.

Software and integration verification

The latest verification passed:

Check	Result
Python test suite	250 tests and 34 subtests passed
Frontend tests	17 passed
Frontend type checking	Passed
Production frontend build	Passed
Missing-data and infeasible-retrofit checks	Passed
Reproducible submission workflow	Previously completed successfully

Regression tests cover critical missing valuation inputs, blocked upgrades, unsupported numerical claims, scenario recalculation, budget clarification, recommendation mapping and streaming-response handling.

Testing exposed and resolved several issues: hidden feasibility reasons, inappropriate package selectors for missing-budget questions, score-improvement questions incorrectly returning baseline explanations, and an LLM describing a final score as a score increase.

Local LLM evaluation

Eight engine-backed scenarios tested score explanations, improvements, heat-pump simulation, budget planning, carbon methodology, unavailable valuation and infeasible retrofits.

Six responses used validated Ollama-generated prose, one used a conversational fallback, and one returned a direct feasibility explanation. Generation attempts took approximately 1.8-4.8 seconds. Numerical token checks found no unsupported figures in the recorded responses.

Human semantic review remains pending. We therefore do not yet claim achievement of the required groundedness, factual-accuracy or unsupported-claim targets.

4. Plan for the remaining time

1. Complete human explanation review: assess relevance, factual accuracy, completeness, feasibility handling and consistency with backend outputs.

2. Strengthen advisor evaluation: expand questions involving missing data, conflicting signals, fuel tradeoffs and interacting retrofits; compare a stronger hosted model if local interpretation remains inconsistent.

3. Improve uncertainty calibration: investigate the below-target property-value change coverage and retain transparent warnings.

4. Rehearse the complete demo: test property selection, assessment, recommendations, budget optimisation, comparison and advisor interaction through the browser.

5. Finalise submission reproducibility: refresh evaluation reports and the submission bundle, verify clean setup instructions, and push all completed source code and scripts to main.

The current prototype supports the integrated synthetic-data demonstration. Remaining work focuses on explanation quality, uncertainty, final browser verification and submission readiness.